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FORECASTING THE EFFECTS OF EXCHANGE RATE FLUCTUATIONS ON THE UNITED STATES' TRADE FLOWS WITH CANADA, MEXICO AND BRAZIL USING MACHINE LEARNING MODELS

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Projeto de Pesquisa apresentado como requisito parcial para obtenção do título de Bacharel em Sistemas de Informação, pelo Curso de Sistemas de Informação da Universidade do Vale do Rio dos Sinos (UNISINOS).

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1 INTRODUCTION

The purpose of this study is to investigate the impact of exchange rate volatility on bilateral trade flows, with particular emphasis on the trading relationship between the U.S. and its main partners, Canada, Mexico and Brazil, which is its largest partner in Latin America (USITC, 2025).

Canada and Mexico trade with the USA through USMCA (United States-Mexico-Canada Agreement). However, Brazil trades extensively without a full-fledged free trading agreement like that (USTR, 2020). The comparison between an advanced and a developing economy will enable us to detect nuances or differences on the pattern of sensitiveness that foreign trade demonstrates when reproducing exchange rate fluctuations, contributing to enhancing the already well-established debate (AUBOIN; RUTA, 2011).

The period analyzed and tested will be from 2000 to 2025, with a combination of historical data and forecasts in the determination of trends and estimations of future effects. This study is also interested in how fine-grained the data will be regarding monthly to semi-annual time slices, so as to accommodate the changes of volatile unpredictable trade flows and exchange rate movements (SHEN et al., 2021).

Apart from an aggregate trade perspective, the study will also offer sectoral analysis to ascertain why different commodities and services may respond asymmetrically to exchange rate fluctuations, offering a better understanding of the heterogeneity of effects among sectors.

The proposed study will use state-of-the-art machine learning models (Random Forests, LightGBM) to forecast the effect of such fluctuations and compare them to traditional statistical methods such as ARIMA in terms of predictive ability (SONG; CHEN, 2024). It will include in its analysis the exports and imports of goods between the U.S. and the chosen countries, using as explanatory independent variables, macroeconomic indicators such as GDP, nominal and real exchange rates, volume indices of production, and inflation (KALLIANIOTIS, 2022). In addition, trade relationships and the exchange rate

dynamics can be seriously affected by outlier events (such as global financial crises and pandemics), therefore the influence of such extreme observations on predictive models will be examined to see how well they hold up in these high-volatility instances due to nonlinear relationships (BATARSEH et al., 2020). This research will follow a structured approach according to CRISP-DM, from business understanding to deployment of the model (MARTÍNEZ-PLUMED et al., 2019).

The primary question to be answered is: "What is the impact of exchange rate variations on export and import flows between the United States and Canada, Mexico, and Brazil, and which forecasting models are best for making these forecasts?"

There will also be additional questions regarding: What are the horizons where predictive power is most informative for assessing the impact of exchange rate movements on international trade, and how does this predictive power vary across different models? Which industries are more sensitive with respect to exchange rate changes and how does this vary between developed countries versus emerging economies? In addition, this study endeavors to explore the asymmetric effects on exports and imports in reaction to currency appreciation and depreciation with a view to finding differences in behavior. We will also study the effect of the depreciation of the dollar on bilateral trade balances, examining how this type of monetary adjustment affects American export prices in foreign markets as well as import prices for American consumers. Moreover, it will compare the forecasting ability of traditional econometric models with that of machine learning models in forecasting non-stationary or strongly fluctuating time series.

The general objective of this study is to assess the relative performance of Random Forest and LightGBM models with respect to ARIMA model in forecasting how continuously changing exchange rates affect bilateral trade flows between the United States, Canada, Mexico, and Brazil with granular level insights about sectoral movement and macroeconomic effects.

The specific objectives are:

a) To pinpoint which sectors are most vulnerable to exchange rate changes;
b) To determine the impact of these swings across different time horizons;
c) To evaluate the predictive power of each model;
d) To examine the effect of macroeconomic factors on the volatility of trade flows.

Ultimately, an end-to-end model-based analysis system for the prediction of trade flows is proposed, serving policymakers and traders for strategic planning. The model comparison will enable the identification of the most adequate methodologies to different context of data and economic scenarios, adding knowledge to the advancement of the literature about predictive modeling in international trade and multivariate time series.

Exchange rate volatility is a key factor that in fact affects international competitiveness, investment decisions, and resource allocation in the world market (RAHMAN et al., 2025). Detailing how exchange rate changes translate into trade flow changes is therefore crucial to designing efficient economic policies and supporting strategic decisions for exporter and importer companies (KIM; LE, 2024).

Accuracy in predicting these parameters with the use of modern machine learning algorithms, when compared to the traditional econometric methods and IT tools developed upon these solutions, become highly relevant as going from minimizing costs over maximizing profits. This accuracy is an essential asset for nations and companies active on the international markets by helping them adapt faster to those changes encountered at a global level (ABIR et al., 2024).

Moreover, beyond its practical contributions, this work is relevant from a methodological point of view, as it relies on the implementation of Data Science best practices, such as ETL process and experiment tracking, to enforce the reproducibility of modeling experiments, and therefore results, in a transparent and consistent manner, thus promoting scientific validity. As emphasized by Vranješ and Niggemann (2024, p. 1–2),

> the iterative process of training, testing, and validating ML models requires empirical evidence to ensure their accuracy and reliability in real-world applications, making experimentation a cornerstone of the ML research methodology, which reinforces the relevance of structuring, documenting, and monitoring the experimental workflow adopted in this study.

In this way, the paper also extends the literature on predictive modeling of international trade, representing a reference framework to scrutinize the ability of trade flows to adjust to exchange rate and macroeconomic interferences.

2 THEORETICAL FRAMEWORK

This section presents the theoretical basis to understanding how exchange volatility impacts international trade flows, and how predictive models are used to assess these dynamics. To ensure these answers are well-founded, we will seek support from the fields of economics and international relations, as well as from the technological field.

2.1 Economic key concepts

Nominal exchange rate: The nominal exchange rate is the relative price of two currencies and plays a major part in determining international competitiveness due to its impact directly on the price of imports and exports (KIM; LE, 2024). As stated by Kallianiotis (2022, p. 46), “Devaluation increases the price of imports and reduces the price of exports; and due to the law of demand, imports are falling and exports are increasing and the trade account is improved”.

REER: Real Effective Exchange Rate (REER) is a broader-based measure that represents the relative price of goods and services of a country with respect to those of other countries in its trading partner, adjusted for differences in prices while accounting for inflation rates (WAHAB, 2024). It thus offers a better understanding and knowledge of an economy’s competitiveness vis-à-vis others since it considers inflation differentials that would, otherwise, either erode or magnify the impact of nominal exchange rate changes (MWANGE et al., 2022).

Export and Import: Imports and exports are key parts of international trade. Imports represent the purchase of foreign goods or services, while exports are the sales of domestic goods or services abroad. These flows are of key importance for a country's trade balance and general economy (NOTZ; ROSENKRANZ, 2021).

HS: The Harmonized Commodity Description and Coding System (HS) is a multifunctional international nomenclature that offers a common language for handling classifications by trade sectors, indispensable for recording trade flows

at more detailed levels (SCHMIDT; STEINGRESS, 2019). The granularity offered by the HS allows an analysis of how product categories react differently to exchange rate changes, yielding very important sector-specific heterogeneities (SELLAMI et al., 2024).

2.2 Models for determining the exchange rate and choosing exchange rate variables: USD/XXX rate and REER

Various theories have been developed to try to explain those factors affecting the determination of exchange rates; it can be inappropriate to rely on one model to predict, as stated by Mwange (2022, p. 71),

> Even though exchange rates can be confusing, practically everyone has to understand them to function in the modern global economy. As the world gets smaller, there is a greater likelihood that we will have to deal with the risks associated with the fact that there are many different currencies in circulation around the globe and that these currencies will have a direct impact on our world. We must be able to evaluate how changes in exchange rates will impact our financial planning, corporate strategies, governmental initiatives, and other life decisions in order to take the necessary action (both financial and otherwise). Examining various exchange rate theories can help us.

The analysis of the nominal exchange rate (USD/XXX) is based on traditional methods in the literature, which involves the link between interest rates, inflation and expectations of devaluation as proposed by the International Fisher Effect (IFE), and the relationship that exists between domestic interest rates, on one hand, and foreign rates on other hand, in accordance with Interest Rate Parity (IRP) which explains how exchange rate is connected to equilibrium among money supply as well as demand for it according to monetary approaches. Another relevant theoretical framework is the balance of payments theory, which considers the exchange rate as being determined by the general conditions of foreign transactions and financial exchanges among countries (MWANGE et al., 2022). An assessment based on the U.S. dollar's nominal exchange rate is conspicuous by its superiority as an anchor currency to inform and shape prices internationally even for trade with third countries that do not include the U.S. directly (GOPINATH et al., 2020). As SPRING-RAGAIN, (2024, p. 4) states,

> Leading currencies, such as the US dollar (USD) and the euro (EUR), play a central role in the global economy due to their extensive use in

> international transactions, foreign exchange reserves and financial markets. Their importance lies in several key aspects: as a store of value, where the world's central banks hold substantial reserves in USD and EUR to stabilize their own currencies and manage economic crises; as a medium of exchange, as the dollar and euro are widely used for international trade, facilitating exchanges between countries; and as a unit of account, as many financial contracts, including international loans, are denominated in USD or EUR, reinforcing their central role.

Thus, the choice of USD/XXX enables a straightforward analysis about the effect of US currency on trade balance.

The use of the Real Effective Exchange Rate (REER) is justified since it adjusts for inflation differentials and trade structures with various partners, hence reflecting the logic of Purchasing Power Parity (PPP) but going beyond it by addressing its failure to account for shifts in trade flows and bilateral relationships, as discussed by Mwange et al. (2022). The REER is a key variable for macroeconomic analysis in open economies and essential for interpreting the movement of exchange rates and their consequences on economic policy. A careful measurement of the REER, using current weights, is required for a better understanding of exchange rates and also their consequences on economic policy (BAYOUMI et al., 2005). This role of the REER is related to its ability to provide crucial information for policymaking, by deeper understanding of exchange rate movements and their effects in the world economy, particularly during crises (BAYOUMI et al., 2005).

Therefore, the combined use of the REER exchange rate and the USD/XXX allows for combining both sets of perspectives, which consider the direct impact of the U.S. dollar and global competitiveness adjusted for inflation, thereby improving the predictive power of the analyses to be performed.

2.3 Trade flows and sectoral sensitivity

The structure of bilateral trade flows is a prerequisite for examining the sensitivity of various sectors to exchange rate changes. The detailed bilateral trade data at the Harmonized System (HS) level allows us to determine where imports and exports go, what their value, volume, weight and thus modeling specific sectoral trade relations. This level of granularity enables the analysis on how changes in exchange rates, particularly those of the dollar, affect volumes of

trade (GOPINATH et al., 2020). Thus, it is possible to determine areas where trade is more vulnerable or stable in the face of exchange rate shocks.

2.3.1 Sectoral impacts of exchange rate fluctuations

Exchange rate fluctuations affect sectors and products differently, and are considered an important determinant of the competitiveness of goods in international markets (NTHEBE; MOSIKARI, 2025).

Considering sectoral specificity is fundamental for accurately estimating the elasticity of export and import demand, especially at the macroeconomic level. The specialized literature shows that the exchange rate has had differentiated impacts on prices across various sectors (GODA et al., 2024).

2.3.2 Commodities vs. manufactured goods vs high technology

Within different product types, such as commodities, manufactured goods, and high-tech items, response to the exchange rate vary, as these products have a particular complexity in the global value chain, in the elasticity of demand, and in their cost dynamics (GODA et al., 2024).

Commodities, for example, are sensitive to exchange rate fluctuations, as they reflect international prices which, in turn, respond directly to variations in the dollar. These factors include production costs and the loss of export competitiveness in a strong currency environment, automatically affecting export incentives and domestic supply in an immediate way (OLIVEIRA et al., 2023). Product complexity and the degree of technological intensity play important roles in determining the sensitivity of sectors to exchange rate fluctuations, with low-complexity sectors in developing countries exhibiting the greatest elasticities in relation to variations in the real effective exchange rate. This is because less sophisticated sectors generally have more flexible production costs and are much more dependent on relative prices to maintain their competitiveness in the international market (GODA et al., 2024).

From an opposite point of view, BERGIN and CORSETTI (2020) claim that industries with differentiated goods, with nominal rigidity and costly entry, are less reactive to the exchange rate than members of industries producing very substitutable goods. This sensitivity is reduced by the nature of imperfect competition character and a dependence on preconditions at the stable macroeconomic level.

2.4 Macroeconomic control variables

The Gross Domestic Product (GDP) and the GDP per capita are among the most commonly employed macroeconomic variables for the modeling of trade flows as they track economic development level and the supply and demand capacity of countries. The GDP per capita, which is GDP divided by population, is especially important for measuring wealth disparities between countries and how these might relate to trade patterns. As noted by Wahab (2024), this variable helps explain how similarity, or dissimilarity, in factor endowments can affect bilateral trade patterns, although it presents potential bias problems, such as the accuracy of population data and the inherent limitations of official statistical data.

Another fundamental variable, conceptually linked to the exchange rate as a dynamic condition, is inflation, which exerts a significant influence on the international competitiveness of products and on the trade balance. For a comprehensive analysis, the examination of the exchange rate regime must also consider external equilibrium in monetary policy, depending on the inflation targeting regime, since flexible rates are effective in correcting external imbalances. Consequently, exchange rate flexibility, a distinctive characteristic of a floating exchange rate regime, can lead to inflationary instability, which, in turn, influences the trade balance (NASIR; VO, 2020).

Fluctuations in the nominal exchange rate also disrupt international negotiations and directly influence production costs and the competitiveness of tradable goods. While depreciation can increase imports of inputs but also make exporting companies more competitive, exchange rate appreciation can result in lower production costs, reducing import prices of inputs (SANTORO; ROLIM, 2023).

Financial and economic crises, which tend to accompany sharp exchange rate movements and financial market turbulence, are also expected to affect trade flows through changes in the supply of and demand for goods and services (ARÉVALO et al., 2022).

2.5 Historical events

The global economy shapes trade flows and exchange-rate patterns as booms and busts change investment, consumption, and internationalization. As mentioned in SPRING-RAGAIN (2024, p. 2):

> Global economic cycles are characterized by alternating phases of expansion and recession, influenced by a multitude of economic, political and social factors. These cycles are often punctuated by major financial crises, such as the Asian crisis of 1997, the global financial crisis of 2008, and more recently, the economic turbulence induced by the COVID-19 pandemic.

2.6 Institutional events

The United States, Canada, and Mexico are home to one of the world's largest free trade areas, initially called the North American Free Trade Agreement (NAFTA), established in 1994, which opened trade and reduced barriers to investment among the three countries. In 2020, NAFTA was replaced by the United States-Mexico-Canada Agreement (USMCA), which preserves regional integration and introduces appropriate structural changes, modernizes trade rules, and includes new regulations (HIGUERA-COTA et al., 2023).

On the other hand, Brazil has long resisted comprehensive trade agreements like the USMCA with the U.S., opting instead to strengthen regional blocs such as MERCOSUR and refusing accession agreements modeled on the U.S. Although the United States wished to expand its agreements with Brazil, the South American country's policy did not align with the requirements of "competitive liberalization" proposed in pacts like the USMCA; instead, Brazil focused on seeking agreements with other Latin American nations (KAT, 2022).

2.7 Other determinants of trade flows

Other factors, beyond those previously mentioned, also impact trade patterns to a great extent. A country's population is one of the most important factors because it indicates market size and potential import demand. The larger the country, the more it can extend its trading network; here countries vary in their market size (SELLAMI et al., 2024). Consequently, a nation's economic capacity, indicated by Gross Domestic Product and national income per capita, crucially affects its involvement in global trade. A strong GDP reflects robust production conditions, which significantly impacts both import competitiveness and the level of exports, while a high per capita income implies greater purchasing power among people, thus promoting demand for higher value-added products and improving both the quantity and quality of national export goods (LI et al., 2024).

Geographical distance is another undeniable determinant of trade, as it represents a natural obstacle that results in transportation and time costs, which ultimately impact the viability of economic relations at an international level. This phenomenon, known as the Gravity Model, has several references in literature, according to which trade between two countries is directly proportional to the size of their economies, measured through variables such as GDP and population, and inversely proportional to the distance that separates them (CONTRERAS, 2022). Related to this topic is logistics infrastructure, which includes the quality and efficiency of ports, airports, highways, and railways. Efficient infrastructure reduces costs and distances, favoring the rapid movement of goods, with a consequent increase in commercial activities. On the other hand, a scarcity of logistics factors can lead to higher transportation costs and inhibit commercial transactions globally (MAO et al., 2023).

Equally important, or even more so, are the connections between states that create a diplomatic environment conducive to trade. Currently, negotiations no longer address only tariff reductions, but also the migration of people, capital and knowledge, as well as cooperation (CARRILLO-MALDONADO; FLORES, 2024). In this sense, conflicts that block these flows can occur. It can also happen that cultural factors, such as similar religious beliefs or a common language, associated with certain countries, result in lower trade costs, combined with

greater trust between partners and, therefore, weakening barriers to entry in trade. A similar culture can correct information asymmetries and influence consumer preferences, thus stimulating trade in goods and services. In some situations, cultural issues are more important for the trade of goods than the quality of the products themselves, especially in goods emotionally linked to cultural aspects (COSTA et al., 2023).

Finally, Global Value Chains (GVCs) are characterized as an international network in which production stages are fragmented and a combination of domestic and imported intermediate inputs is used to produce final goods and services. Countries specialize in different activities, supplying intermediate inputs that are used to produce both for export and for the domestic market (GAULIER et al., 2020). Contemporary trade patterns are profoundly affected by increasing global integration and all the phenomena associated with it, such as productive globalization. In particular, the production of advanced high-technology goods through international specialization and the fragmentation of production stages has intensified trade flows and increased interdependence among participating economies (AKMAN et al., 2020).

2.8 Time series forecasting models

The goal of time series analysis is to model and predict the behavior of sequential data, as well as to discover temporal dependencies or patterns. The conventional approach, reflected by econometric models such as ARIMA, assumes that the problem is linear and that the data are stationary so that statistical patterns remain stable over time and structural relationships can be modeled. Preprocessing techniques, such as differencing, are used to make time series data stationary (ABIR et al., 2024).

Time series analysis is considered basic in the understanding of economic aspects, particularly regarding international trade and networks. It allows the analysis of data at regular intervals in order to identify trends, cycles and patterns that will help to explain the evolution of commercial behavior and the effects of macro events, providing information on past activities, economical dynamics and future performance (YAZAWA, 2023). In this sense, the ability to analyze traffic

aims to detect and predict these common temporal patterns in order to make strategic decisions at the levels of international trade and finance (ARÉVALO et al., 2022).

2.8.1 ARIMA model

AutoRegressive Integrated Moving Average (ARIMA) model proposed by Box and Jenkins in the 1970s, is regarded as one of the classic statistical tools for time series analysis and forecasting applications, especially in the finance field (LIN et al., 2020). The model is particularly suitable for the series which have linear trend, stationarity and no seasonality or can be transformed to satisfy these requirements (HE et al., 2020).

This model has three main parameters: “p,” which is the order of autoregression, “d” as the degree of differencing, such that stationarity is reached, and “q”, as the moving average order, that allows capturing internal dependencies or removing trends or seasonality in time series (CHATTERJEE et al., 2024).

2.8.2 Smoothing and decomposition models

Exponential smoothing models are used for forecasting time series, using local averaging with weighting of past data. In this case, the data have an exponentially diminishing weight as they get older; indeed, any such model will tend to give more importance to more recent events than historical data. This approach is especially favorable when analyzing series with trends and seasonality since exponential smoothing can be modified to accurately model these aspects (SHEN et al., 2021).

Time series decomposition differs by dissecting the data into its structural components, e.g., trend, seasonality, cycle, and error, eliminating or minimizing noise, making the time series more suitable for forecasting. This decomposition allows each part to be modeled separately and then recombined to produce a final forecast (HE et al., 2020).

For the case at hand, such models are instrumental for deciphering complex interactions in economic and financial systems, reporting dynamic responses to shocks and their effect durations (ARÉVALO et al., 2022).

2.8.3 Stationarity tests

Stationarity is a fundamental subject when applying time-series models, such as ARIMA. The differencing (parameter "d") is used where data demonstrate non-stationarity to transform the series to make it conducive for ARIMA fitting (CHATTERJEE et al., 2024). For the adequacy of time series models, formal tests are performed, such as Dickey-Fuller test, to check the unit root and order of integration process of the series that served as basis for further analysis, such as co-integration (ARÉVALO et al., 2022).

Non-stationarity of time series, such as trend and seasonality, severely damages the accuracy of models like ARIMA. Hence, the removal of non-stationarity is an important prerequisite for the application of several time-series models like ARIMA, and its elimination or alleviation has been pursued in advanced approaches to improve forecasting performance (LIN et al., 2020).

2.9 Machine learning in forecasting

Recent advances in machine learning have made possible the construction of more resilient and flexible models to forecast time series, which are capable of dealing with non-linearity and complexity that characterize several economic phenomena, surpassing the shortcomings of traditional statistical approaches (CHINN et al., 2023). Machine learning algorithms, like Random Forest and LightGBM, are chosen for this study due to their ability to capture non-linearity and complex relationships within the data without imposing stringent prior assumptions on data distribution or stationarity conditions (CHATTERJEE et al., 2024).

While Random Forest and LightGBM were not specifically developed for capturing temporal dependency, they are extensively used in time series prediction tasks, due to their flexible ability to model complex relationships and

handle high dimensional data. In these applications, time can be combined by feature engineering using the calendar information such as year and month as well as transformation based on the series itself (ZHANG; JIANG, 2025).

2.9.1 Random Forest

Random Forest is one of the most versatile and widely used machine learning algorithms for handling complex data. It is non-parametric, that is, it can deal with different types of responses and predictors over scales and distributions, making it appropriate for high-dimensional setting and complex data. Its power becomes evident from the scale invariance of its frequency-based functions and the fact that its prediction error remains stable even with an increase in the number of trees (HU; SZYMCZAK, 2023). Hence it helps mitigate the overfitting problem present in isolated decision trees since this model works by creating multiple decision trees at training and outputting the class that is the mode of classes (classification) or mean prediction result (regression) of the individual trees (CHATTERJEE et al., 2024). This ensemble learning technique by combining output from many trees reduces variance and bias, maximizing predictive accuracy (LING; WANG, 2024).

What is more, Random Forest has great practical advantages for applications in economics and business. It is moderately resistant to abnormal points, can cope with multicollinearity between the predictors as well, and is able to generate variable importance measure which further improve applied interpretation. These traits can make the Random Forest a good choice for balancing accuracy and interpretability on empirical scenarios (SALMAN et al., 2024).

Nevertheless, it is important to recognize some of the methodological weaknesses in the model that were raised by SELLAMI et al. (2024, p. 4):

However, while Random Forest models are well suited for analyzing trade data, they have limitations in extrapolation. This inability to extrapolate effectively can hinder their accuracy in predicting future trade flows over time. Furthermore,

in general, deep learning regression models tend to outperform Random Forest models.

2.9.2 LightGBM

Otherwise, LightGBM is a gradient boosting framework that shows efficiency and accuracy on large data sets while also being optimized for both training and memory usages (CHATTERJEE; et al.; 2024). Its architecture can make use of training acceleration techniques and reduce memory usage, which would make it convenient for high-dimensional sparse scenarios. Such innovation significantly contributes to model efficiency when carrying out machine learning duties (GUO et al., 2024).

Gradient boosting models have been designed to be efficient, and it is considered as an efficient tool for a machine learning problem where computation is a concern and data sizes are very large though some models may behave slowly under huge data size (ENKHBAYAR, ŚLEPACZUK, 2025).

In real-life scenarios, incomplete data is common, and as demonstrated in studies, techniques such as oversampling with LightGBM manage this data and address class imbalance (GOPINATH et al., 2021).

While it comes with the advantages of LightGBM, we can acknowledge that exactly interpreting LightGBM still is a difficult task due to its naturally high complexity in tree boosting models. Thus, although the model may result in highly accurate predictions, it may do this less transparently than simple models, which is pertinent when explainability is vital (GOPINATH et al., 2021).

Further, the models like LightGBM are also overly sensitive to overfitting, which requires precise adjustments of numerous hyperparameters and careful regularization. The optimization of these hyperparameters, which can be very computationally demanding, is crucial for avoiding overfitting and promoting the model learning to generalize well (SAIED et al., 2013).

Compared with some other models, which can be so complex that they are not interpretable, LightGBM is usually in a good place on this trade-off. This

model is known for its high accuracy as well as its interpretability, making it an effective option when both predictive accuracy and model understanding are important (GUO et al., 2024).

2.9.3 Comparison between traditional approaches and ML

The main difference between machine learning models, like LightGBM and Random Forest, is that they are intrinsically capable of modeling non-linear relationships or complex interactions in the data. Unlike econometric approaches that usually impose strong assumptions on the properties and distribution of data, machine learning models rely much less on premises and adjust more easily to changing data patterns (SELLAMI et al., 2024). This flexibility makes them particularly appropriate for the prediction of complex phenomena, like exchange rates and flows trade, overcoming the constraints and difficulties inherent in traditional methods that do not always reflect the dynamics underlying data (GUO et al., 2024).

Therefore, a main difference between machine learning and econometric models is that, while in the former the data speaks for itself regarding patterns, in the latter there is a need to make hypotheses based on predetermined model specifications (SILVA et al., 2024).

On the other hand, in terms of data interpretation, available machine learning models are consistently described as “black boxes”, making it difficult to grasp the justification for their predictions compared with traditional econometric models that are a more transparent and built on explicit economic theory (ABOUZAID; BOUSSEDRA, 2025).

2.9.4 Hyperparameters and optimization

Although the "black box" nature of machine learning models presents challenges, interpretability techniques, e.g., SHAP-based methods, are starting to close this divide. These methods, inspired by cooperative game theory, allow for more interpretable examination of how variables influence predictions of the complex modeling task by decomposing predictions into feature contributions and

thereby expose non-linear structures and interactions that might have been hidden otherwise (ZHANG; JIANG, 2025). SHAP is known to be very powerful in understanding the importance of each input feature in a model's prediction about a certain case, and in providing an index representing the contribution of each variable to the output value of the model. This capability to assess the relative importance of each feature for a given prediction improves interpretability and trust in machine learning models, even for highly complex models (GUO et al., 2024).

The use of these methods provides an opportunity for policymakers not only to follow high-fidelity forecast predictions but also to understand the causal reasons behind such forecasts, thus making better and more informed strategic decisions (ZHANG; JIANG, 2025).

However, identifying the essential hyperparameters for model performance is a major challenge in which slight changes can result in large differences in predictive accuracy (ZHANG; JIANG, 2025).

Thus, the comparative application of Random Forest and LightGBM in this work gives options to evaluate their predictability on trade flows under a fluctuating exchange rate as a provision to traditional methodologies. The hands-on application of these methods will include extensive exposure to computational tools, such as Python and its statistical libraries, data integration tools via SQL, visualization resources on GitHub for reproducibility with versioning adopted in this study.

2.10 Technologies and tools

In this context, we provide an overview on the main technologies and platforms used to implement and validate the predicted models. This section is devoted to the description of both the libraries used, the hardware and software environments in use as well as how we handled issues related to scaling up and reducing the computational cost of the analysis. The choice of these tools is to maximize the pipeline, including data collection, preprocessing and modeling, evaluation, and result deployment.

2.10.1 Python for data analysis

Python gained a lot of attention in the data analysis and machine learning world because of its huge collection of libraries and frameworks that are highly needed for designing and testing predictive algorithms (ISLAM; RASHID, 2024).

Python has a robust ecosystem of data manipulation and processing libraries like Pandas and NumPy. These libraries play an essential role in organizing, cleaning and transforming large amounts of data, which then allows for manipulation from a mathematical/statistical point of view and for exploration of relationships and patterns within structured datasets (ATOCHERO; LEDO-DÍAZ, 2024).

The Statsmodels library is heavily used for the examination of statistical and econometric models, like ARIMA. This library is developed specifically for time series analysis and modeling which provides relevant functionality to implement ARIMA as well as exponential smoothing models (HEDAYATI; ASL, 2023).

In the machine learning space, Scikit-learn and LightGBM for modeling and high-performance boosting methods are used. The latter emerges as an important factor in the trade-off between computational efficiency and predictive accuracy, and it is a very adaptive alternative in order to fit complex prediction models for practical applications (SALVADOR, 2024). Some of the more traditional search methods, Grid Search and Random Search, are also present in Scikit-learn and they form a strong part of model tuning techniques to be used. Grid Search performs a systematic search over hyperparameter configurations, whereas Random Search samples combinations randomly across the search space (CHOWDHURY et al., 2024).

Other significant library is Optuna, a hyperparameter optimization framework that stands out for being algorithm-agnostic and allowing efficient searches of more complex search spaces and helping to achieve models with better performance (LING; WANG, 2024). Optuna hyperparameter optimization is essential to improve Random Forest and LightGBM models guaranteeing fine

tuning that captures the complex interactions of exchange rate variations (JABBAR; JALIL, 2024).

These tools together with visualizing libraries like Matplotlib and Seaborn are essential for observing patterns as well. Moreover, we can visualize the results in high quality through them in machine learning studies. Seaborn, based on Matplotlib, is used to analyze and interpret data, and Matplotlib is needed for basic data visualization applications. The use of Jupyter Notebook environments, which allows the insertion of data visualization, highlights the relevance given to these tools for presenting results (ZAIM et al., 2022).

2.10.2 SQL for data integration

One of the most important steps in the consolidation of information is to merge disparate pieces of data, this requires cleaning, normalization and resolving inconsistencies among data sources for coherence and compatibility between datasets. Tools like SQL (Structured Query Language) for data retrieval, analysis and integration with data-warehouse repositories for efficient access are critical. This method is essential in achieving uniform and easily accessible datasets, which are crucial for constructing predictive models, especially in the machine-learning environment (REMIL et al., 2024). Furthermore, the use of SQL allows for the implementation of complex queries as well as stored procedures, which is ideal for data accessing and manipulation in a large-scale scenario (ISLAM 2024).

2.10.3 Reproducibility and versioning

For the purpose of transparent research, being able to replicate and reproduce results is essential in order for others to be capable of verifying and expanding on the work. In this perspective, version control systems like Git and code hosting platforms like GitHub can be very much required to track down in a systematic manner the changes of data and code, and to manage different versions of the project (KAYIT; ISMAIL, 2025).

In addition, a thorough documentation of all steps from data acquisition and preprocessing to model calibration and evaluation is required in order to provide the basis to accurately repeat the experiments (SEMMELROCK et al., 2024).

Finally, containerization strategies, e.g., Docker, are essential for guaranteeing the uniformity of the code execution environment. Because the computational environment is “frozen” into an “image” in the sense of a so-called container that can be copied from machine to machine, Docker ensures that code and its execution environment are homogenous and not impaired by irrelevant inconsistencies originating from slightly different software configurations or overall systems, thereby also contributing to computational reproducibility (PEIKERT; BRANDMAIER 2021).

3 METHODOLOGY

In this research, the Cross-Industry Standard Process for Data Mining (CRISP-DM) was used to predict the impacts of changes in exchange rates on trade flows between United States and Canada, Mexico, and Brazil. This decision was motivated by the need to connect with macroeconomic database, time series data and predictive models.

The methodology of CRISP-DM is widely recognized as one of the most commonly used methodologies in data mining projects, which organizes the project into well-defined phases. That structure guarantees the quality and transparency of the models at all stages, from business understanding to deployment, enabling the generation of insights and implementation analytical solutions (TRIPATHI et al., 2021).

The process consists of six phases that will be the focus of this chapter: Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation and Deployment (WIRTH; HIPP, 2000).

3.1 Business understanding

This is the first phase of the CRISP-DM methodology, where the project's goals are set and the required actions are determined. At this stage, requirements are transformed to a definition of the data mining problem and project plans defining activities for several subsequent steps (SHIMAOKA et al., 2024).

This stage also covers enlightenment in the economic and institutional environment, identification of external forces like exchange rate volatility, global macroeconomic shock and structural asymmetry. These factors are very important because they affect both analytical models and the interpretation of the results, and they should be included in policy making and whenever we evaluate global impact (NGALAWA; KUTU, 2017).

Also, at this stage, the project’s success criteria are defined. These criteria will be used as the comparison standard for assessing both the models’ outcomes and the practical usefulness of the forecasts produced by them (MANDOLINI et al., 2024).

3.1.1 Business objectives and stakeholders

To ensure that the results of the forecasting model are directly useable for strategic business decisions, a profound knowledge of the business objectives is necessary in order to be able to adjust expectations. This initial comprehension covers from theoretically mapping the influence on trade flows and currency exchange variations to uncovering relevant actors, including potential users of the model such as stakeholders, policy and decision makers (MOTTA et al., 2021). This will help in ensuring that the solution developed gives insights for actions, supporting business decisions, so the organization can achieve better outcome (SATRI et al., 2023).

Methodologically, business understanding in the first alignment phase is important for calling into question what aspects of the economy should be given greater depth of analysis later on. The above features encompass, but are not limited to, the sectoral sensitivity analysis and export-import variation, as well as the effect of macroeconomic incentives on trade flow decisions, an important consideration for shaping our understanding of trading patterns and policymaking (BATARSEH et al., 2020).

3.1.2 Scope, core problem and theoretical grounding

In this study, the Business Understanding phase consists in determining which are the most important macroeconomic variables, defining performance metrics to evaluate how well predictive models perform and very importantly, clearly stating the core problem to be solved. In this respect, the global purpose of the project is to assess the accuracy of Random Forest and LightGBM compared to ARIMA forecast when predicting trade flows between the United States and Canada, Mexico, and Brazil, based on the currency fluctuations rule, from the year 2000 to 2025, that dictated the initial definition of project features.

The choice of these variables and metrics is not arbitrary, it comes directly from the prior review of theory and what is known about traditional models such as ARIMA being limited for complex macroeconomic time series, and the potential of machine learning methods to model nonlinear relationships (LIN et

al., 2020). The project will be guided by widely accepted predictive error indicators, which will be specified in the Evaluation phase.

3.1.3 Linking business understanding to research objectives

Furthermore, Business Understanding should structure the remaining stages of the methodology in order to face specific research objectives like the following: identifying the sectors most vulnerable to exchange rate changes; analyzing how these effects vary over different time horizons; evaluating the relative predictive power of each model considered, and examining the role of macroeconomic factors in the volatility of trade flows.

Therefore, the first phase is Business Understanding, which can transform the objectives of this work into concrete requirements. This is followed by Data Understanding, where the data is explored and assessed to determine its context and quality for the expected analysis.

3.2 Data understanding

The Data Understanding phase, as it appears in methodologies like CRISP-DM, is the time when data is described, explored, and its quality and availability are checked. This process will become the basis for a plan for future transformations. For recognizing patterns, discovering limitations and hypothesis-making, the raw data is being collected and analyzed. This phase is often exploratory and concerned with assessing data quality, which underpins the success of experiments and the trustworthiness of information (RIESS, 2022). The assessment of the completeness, consistency and precision of datasets and any potential biases in them is important for subsequent data science tasks. As pointed out by MARTÍNEZ et al. (2021), the lack of harmonized diagnoses and quality controls, which leads to errors, is a reminder that knowing the data's availability, representativeness, and limitations is an important step for successful projects.

3.2.1 Data sources and scope

The kind of data for analysis and prediction models is usually composed of historical exchange rate data taken from worldwide financial markets or central bank respondents, as well as macroeconomic indicators like interest, inflation, GDP growth, trade balance, and foreign direct investment obtained from sources such as the World Bank (WB), International Monetary Fund (IMF) or national statistical bureaus (ABIR et al., 2024).

In this analysis, these will encompass historical series of USD/CAD, USD/MXN, and USD/BRL exchange rates taken from central banks and the Federal Reserve (FED), bilateral trade flows retrieved from databases such as UN Comtrade or U.S. Census Bureau, general macroeconomic indicators from the World Bank or Organization for Economic Co-operation and Development (OECD) as well as sectoral data sources such as commodities, manufactured goods, and high tech goods useful to capture the sectoral asymmetry of trade flows.

3.2.2 Frequency, length and temporal coverage

The frequency of this data ranges from monthly to quarterly, depending on the existence and consistency of the series. This flexibility is essential for adjusting the model to specific time horizons of analysis and enables an understanding both of short-term effects and long-term structural activities. Such data sets have often had a length of more than a decade, enabling longitudinal analyses and the identification of macroeconomic events that influence exchange rate dynamics and trade flows. (ABOUZAID; BOUSSEDRA, 2025).

3.2.3 Tools, integration and preliminary transformations

Data manipulation and integration for analysis typically occurs using the Python programming language with packages including Pandas and NumPy used for data loading and processing (BATARSEH et al., 2020). Although specific transformation and cleaning steps are formalized in the subsequent phase of the method, at this step we identify the requirement for integration of structure,

standardization on variables and verification for compatibility between different data sources.

3.2.4 Exploratory statistics and quality assessment

In the early stage of data exploration, basic descriptive statistics such as the mean, median and standard deviation are computed to understand distribution patterns and for better investigation of relationships between variables (LI et al., 2025). This stage provides initial reports and preliminary insights regarding the appropriateness and quality of the data. The identification of data quality at this point is especially important since one needs to implement cleaning and transformation mechanisms to make the data accurately reflect the macroeconomic and sectorial aspects related to the problem under consideration, and to guarantee that the result of any analysis will be reliable. Therefore, an evaluation is made that guides the following data cleaning and transformation to have enough information to be used in analysis processes (VERSTRAETE et al., 2021).

Once Data Understanding is done, we enter the next phase which is Data Preparation: organizing database in order to have consistent, complete and data fit for the purpose of models that are going to be used.

3.3 Data preparation

After the data has been collected and made sense, the Data Preparation phase aims to ensure the quality and suitability of the raw data, guaranteeing that it is properly cleaned, transformed, and formatted. This includes rectifying errors and inconsistencies, treating missing values, re-coding data types to a standard format and normalization, standardization or coding of the variables for their appropriate use in model building actions (BÂRÃ et al., 2024).

Models like ARIMA, Random Forest and LightGBM, which this paper is to address, require careful data treatment to ensure optimized performance and valid predictions. As mentioned in (MA et al., 2024), the adequacy and quality of input data are essential indicators of the success and reliability of model results.

Data poor quality obscures the attainment of technical goals and meticulous preparation of data by maintaining cleanliness, consistency and format ensure the validity and applicability of inferences as well as predictive insights rendered by the model.

This is also considered to be one of the most important steps since bad data in means bad data out, leading to poor model performance and erroneous conclusions, regardless of algorithmic sophistication (MA et al., 2024).

3.3.1 Variable screening and selection

Screening variables is an important aspect in the process of data preparation, aiming to identify and retain only relevant features that contribute to the model's prediction and performance. (MA et al., 2024).

Macroeconomic variables, such as exchange rates, GDP, interest rates, exports and imports, and trade sectors are important factors to be considered in studies about economic dynamics and international trade. The use of these variables is methodologically well-founded in ample economic literature that supports a strong and significant relationship between these factors and the specialization patterns and international trade flows (SOUSA; VERÍSSIMO, 2021).

3.3.2 Data sources, integration, frequency conversion and cleaning

Integration of several databases is necessary to create a unified dataset that contains consistent and complete information needed for modeling, which involves integrating disparate datasets into a single framework for analysis (SOH et al., 2021).

Moreover, the data frequency conversion is also essential for the adjustment of models like ARIMA, which may require a certain temporal consistency for optimization of predictions (BRYAN et al., 2023).

At the same time, data standardization is essential to make analysis results comparable and valid, particularly in studies that examine different regions or

periods. The data heterogeneity in different economies may hinder model relevance and obtaining consistent results (ABIR et al., 2025).

Data integration and manipulation, implemented with the support of Python libraries, are important steps for organizing and validating information. This includes date control, the verification of data integrity and consistency as well as temporal alignment for variables (MANDADAPU 2024). Use of data transformation methods, e.g., dummy coding for nominal predictors is also important to prepare the dataset for machine learning algorithms (SILVA et al., 2024).

Then, missing values are generally dealt with using both the removal of null values and imputation, where methods like that implemented by the k-Nearest Neighbors (k-NN) algorithm may be used for improving the estimations (BRYAN et al., 2023).

3.3.3 Outlier detection and treatment

Outlier detection and processing, which consists of identifying atypical observations, is significant as these observations without treatment, may mislead statistical results and prediction accuracy (TAJMOUATI et al., 2022).

The identification of outliers may be carried out by using descriptive statistical methods that summarize information on attributes, including the use of the quartile range, which detects values that fall far away from the upper and lower quartiles, and by visual inspection of graphs, because this is a useful technique for identifying odd observations within data (SUDJIANTO; ZHANG 2024).

Nevertheless, automatic exclusion of outlier data is generally not recommended, because the decision to exclude these anomalous observations should always be based on consideration of their collection process and the impact they have on predictive models, along with appropriate validation results to prevent misclassifications. Sometimes it may be more reasonable to include outliers that will be helpful for checking the model robustness, rather than claiming about invalid data (DIAS et al., 2020).

3.3.4 Transformations: differentiation, normalization and encoding

Differentiation is a widely used method in time series analysis, especially in econometric models where trends and seasonality have to be eliminated, thereby stabilizing the series mean and addressing non-stationarity (LIN et al., 2020).

However, in machine learning algorithms, feature normalization and standardization, through techniques such as min-max scaling, is employed to adjust variables to a common scale, typically between zero and one, preserving the relationships inherent in the original data and preventing larger magnitude values to swing towards dominating interpretation (VOLOSHKINA et al.,2024; HSIEH et al., 2024).

In addition, the qualitative categorical variables are binarized to avoid outliers and allow for a good interpretation of such attributes by the machine learning models (GONZÁLEZ et al., 2022).

3.3.5 Feature engineering and temporal features

Data Preparation includes feature engineering process for building the performance and predictability of ML models, generating and transforming new variables from existing ones, which will make data more meaningful to model (CHOWDHURY et al., 2024).

This operation is necessary to extract hidden and non-trivial relationships not directly explained by the original variables, to enhance predictive performance of machine learning techniques (ABIR et al., 2025).

Besides, a subset of time series transformations is introduced, in particular lags, moving averages and percentage variations, which aid in grasping temporal behavior that is indispensable for econometric and machine learning models, while preserving the initial sequential structure of the series. The generation of lags is worth focusing as the order of lag highly contributes to increase precision of the forecast (LIN et al., 2020).

3.3.6 Temporal alignment and final dataset readiness

To make sure the time sequence is consistent, an important part of Data Preparation. Since time series are based on the temporal component, fine-grained manipulation and alignment of temporal data have to be performed correctly in order to avoid wrong lags as well as ensure that causality and predictive relationships that the models have learned about still hold valid (VERSTRAETE et al., 2021).

Once these steps are completed, a clean, structured dataset is obtained, ready for the Modeling phase, in which the algorithms will be adjusted, compared, and evaluated according to the study's objectives.

3.4 Modeling

The process flow for Modeling, as an immediate continuation of Data Preparation, is next after preparing and engineering features, as outlined in the CRISP-DM methodology. At this stage, techniques like statistical tests and machine learning algorithms are executed iteratively to construct, select and refine predictive models used to detect patterns in the data (CHOWDHURY et al., 2024; MA et al., 2024).

In this paper, we will utilize and compare both econometric models like ARIMA and machine learning models such as Random Forest and LightGBM to take advantage of their time series forecasting ability.

3.4.1 General overview of the modeling approach

In general, the modeling approach in time series prediction problems involves a sequence of steps. Firstly, (i) the definition and preparation of features for assembling a supervised model, including variable transformation and data cleaning. Right away, (ii) finding a prediction window is essential to the predictive goal of the model. The (iii) division of the data into training and test sets is a step that aims at assessing how well the model's performance on unseen data, simulating a real-world context. Afterwards, comes the phase of (iv) model tuning,

whereby hyper-parameters are adjusted, e.g., using of cross-validation ensuring that the derived models generalize well. Then, the (v) comparative evaluation is compared in terms of its strategy impact and economic contributions, based on metrics. Finally, (vi) explainability analysis of the model is appreciated to gain higher performance and confidence (SILVA; TIMO, 2022).

Avoiding data leakage, the so-called "contamination zone" effect, is important because inconsistency between the training and testing phases can lead to erroneous assessments of model performance (REMIL et al., 2024).

3.4.2 Data format for modeling

The way to define time series models as inputs is varied, and ARIMA models are widely used for univariate time series. Furthermore, models like SARIMA, a variation of ARIMA, can also contain mechanisms to deal with non-stationary data, such as trends and seasonality (PORTO, 2022).

For Random Forest and LightGBM, both of which are models based on decision trees, the data input can be multivariate and does not require time series stationarity. This flexibility allows them to more easily incorporate several macroeconomic lagged variables (BOTZ et al., 2025). This flexibility enables the incorporation of features derived through feature engineering, e.g., lags, which add time delays to allow models to capture short and medium-term dependencies, moving averages, and calendar features, all of which are pivotal for capturing temporal and seasonal dynamics in the data (ZANOTTI, 2025).

3.4.3 ARIMA model: definition and justification

The AutoRegressive Integrated Moving Average (ARIMA) model has been used as a benchmark method for time series analysis, such as financial and economic time series. Its approach involves the removal of non-stationarity, primarily through differentiation, to enable modeling the time series (LIN et al., 2020).

Its parameterization (p, d, q) models autoregressive dependencies (p), the level of differentiation required to remove trends (d), and the moving average component (q) (RIZVI, 2024). ARIMA has been employed as a fundamental baseline model for comparative forecasting performance studies, and in certain applications, it can even emerge as the best approach for short-term forecasts, serving as an effective baseline for the evaluation of more complex models (BOTZ et al., 2025).

3.4.4 Specific procedures for ARIMA

For each single-variable time series, the standard modeling practice starts with a stationarity test, for which the Augmented Dickey-Fuller (ADF) test is often applied, to determine the order of differentiation (d) of the series. Examination of the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots are useful to suggest the appropriate parameters (p) and (q) for the model. These visual analyses allow for the reduction of the parameter search space (REICHERT; SOUZA, 2020).

The final model selection is then performed using information criteria such as the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC). The goal is to identify the minimum values of these criteria to detect the most parsimonious and best-fitted models. Finally, a rigorous diagnostic analysis of the residuals is crucial to confirm the absence of heteroscedasticity and autocorrelation, which are White Noise properties, thereby validating the model’s assumptions and ensuring its adequacy (REICHERT; SOUZA, 2020).

The backtesting process for models such as ARIMA can occur retrospectively using a sliding window approach. Models are trained on historical data up to a point in time (t0), and predictions are then generated for the following period (t0+1). This process is repeated sequentially, thereby dividing the time series into several training and testing windows, which allows for the comparison between predictions and actual values and the evaluation of the model's predictive stability over time (BOTZ et al., 2025).

3.4.5 Preparation and modeling for Random Forest

Random Forest is implemented using an ensemble approach which aggregates multiple decision trees over the supervised matrix, as it is applied in prediction tasks (ABIR et al., 2025).

Unlike ARIMA, it performs well for non-linear relationships and is capable of capturing complex interactions between predictors without assuming the underlying data to be stationary (CHATTERJEE et al., 2024).

The aforementioned algorithm is tolerant of various data types and less sensitive to outliers and multicollinearity, which is an advantage for dealing with macroeconomic and financial data (LING; WANG, 2024; SALMAN et al., 2024).

3.4.6 Preparation and modeling for LightGBM

Similar to Random Forest, Light Gradient Boosting Machine (LightGBM) is a powerful time series forecasting algorithm. This specific algorithm is known for its speed and accuracy, often demonstrating superior performance in processing large, high-dimensional datasets (ZANOTTI, 2025). It is also based on a supervised matrix, but it requires caution in the optimization of hyperparameters and the treatment of missing values (MORTEZANEJAD; WANG, 2025).

This model is particularly strong when it comes to dealing with complex interactions between features and uses a histogram-based method to construct trees. By binning the training data, this approach significantly decreases memory consumption and speeds up training. For this reason, LightGBM is highly efficient for data with many features and can effectively learn complex patterns, outperforming other algorithms in both accuracy and execution time (MORTEZANEJAD; WANG, 2025).

3.4.7 Precautions with machine learning models

Nevertheless, the employment of well-tuned machine learning models for financial market prediction, though appealing, should be treated with caution due to uncertainty about how well such models can generalize across diverse market

conditions and the need for thorough sensitivity analyses to validate model stability (WANG; XIE, 2024).

Besides, the risk of overfitting should also be considered to ensure effectiveness in real-world scenarios and durability of the algorithms across different historical periods (WANG; XIE, 2024).

3.4.8 Temporal validation and evaluation

Temporal validation is key in time series analysis because it maintains the temporal order of the data, avoids information leakage from future to past, and ensures model evaluation under realistic forecasting settings (MORTEZANEJAD; WANG, 2025).

Techniques like the rolling window and the expanding window are employed. With the rolling window, the model is trained on a fixed period and this window slides over time. The expanding window, however, adds new data to the training set in each iteration, expanding the available historical data horizon (ZANOTTI, 2025).

3.4.9 Training, validation and test split

To minimize problems of improper evaluation and over-generalization, time series should be strictly split according to chronological order (KHAN et al., 2024).

Typically, approximately 80% of the series is employed for model training, while the remaining 20% is used for prediction and performance comparison. This preserves the temporal structure and prevents access to future data during training, thereby mimicking the natural flow of time and consequently avoiding data leakage (KHAN et al., 2024; SAFAROV et al., 2025). This approach is adopted in this study.

ARIMA models are univariate and will therefore straightforwardly process one time series. Data manipulation and preparation, including the differencing operation to achieve stationarity, are done intrinsically at the level of the series

itself; that is, no external shifts or alterations are made for target vectors, as in other approaches (BOTZ et al., 2025).

Because of this structure, in addition to training and validation, this approach employs backtesting strategies based on sliding window methods. This is often referred to as time series cross-validation (TS-CV), which enables iterative evaluation of model performance, thereby ensuring that both training and test data preserve chronological order during validations (ABIR et al., 2024).

3.4.10 Hyperparameter optimization

In Machine Learning model development, hyperparameter tuning is essential for maximizing performance, with various techniques being applied (VRANJEŠ; NIGGEMANN, 2024).

The most common methodologies include: Grid Search, an exhaustive search approach that evaluates all predefined configurations; Random Search, which selects configurations randomly and is often more efficient, finding similar performance while shortening the computational time; and Optuna, which involves training a probabilistic model relating hyperparameters to the model's performance metrics. Each methodology has its unique characteristics that impact the efficiency and scope of the hyperparameter search space (RASCHKA et al., 2020; LING; WANG, 2024).

Given the modeling approach, as part of the CRISP-DM process flow, we move to Evaluation, which allows models to be compared based on their accuracy and practicality of use.

3.5 Evaluation

In the CRISP-DM methodology, the evaluation phase is essential to verify that the models developed not only meet the business objectives previously defined in the Business Understanding phase, but are also sufficiently tuned for deployment (SATRI et al., 2023).

Model validation is achieved by scrutinizing the performance, interpretability, and robustness metrics of a model to confirm its credibility and utility in decision-making (SUDJIANTO; ZHANG, 2024).

3.5.1 Statistical and economic evaluation metrics

To compare the models, the assessment of predictive accuracy and practical relevance requires the use of robust statistical metrics, such as the Root Mean Squared Error (RMSE), the Mean Absolute Error (MAE), and the Mean Absolute Percentage Error (MAPE), all of which are frequently employed to gauge prediction accuracy (AYITEY et al., 2023).

Besides statistical metrics, economic metrics such as simulated investment profit, net and gross financial returns, operating costs, and risk ratios, e.g., Sharpe and Sortino ratios, are crucial for evaluating the practical applicability and financial viability of the models in real-world market scenarios, thereby providing a comprehensive view of performance (MACHADO et al., 2020).

Performance measures are necessary for the purpose of comparing machine learning models to each other and to traditional approaches like ARIMA to determine which model is most effective for a specific application. These evaluations are essential to quantify the value added by complex models over baseline methods and also validate the dependability and generalizability of the model (UDDIN et al., 2025).

3.5.2 Interpretation and explainability of models

Although machine learning-based models may provide superior performance, their interpretability is often difficult to achieve, particularly with financial applications where the clarity of the decision-making process and transparency are a necessity for decision-making and regulatory compliance (SUDJIANTO; ZHANG, 2024).

To circumvent this opacity, interpretability methods such as feature importance analysis and SHAP-based explainers are applied. These methods

elucidate the contribution of each input variable to the model's predictions, which facilitates a deeper understanding of its decision-making process and the individual impact of features on the outcomes (GUO et al., 2024).

This enables policymakers and financial analysts to understand not only “what” is predicted, but also “why” certain predictions are generated. Overcoming the opacity of “black boxes” is crucial for cultivating trust and the adoption of these predictive tools (ABOUZAID; BOUSSEDRA, 2025). In addition, the model’s interpretability is crucial for users such as policymakers and investors to understand its forecasting logic and make informed decisions. This indirectly validates the coherence of its relationships with economic indicators (KHAN et al., 2024).

Moreover, explainability is crucial to identify potential biases in models and to ensure compliance with regulatory requirements, particularly in sensitive sectors, e.g., the finance sector (SUDJIANTO; ZHANG, 2024). This ability to account for the working of the model, including a detailed explanation of how economic variables influence outcomes, is also essential for supporting its predictive ability and helps ensure that it reflects a sensible understanding of the underlying economic principles (KHAN et al., 2024).

3.5.3 Robustness tests and sensitivity analysis

Robustness tests and sensitivity analysis are the methods used to check the validity and performance of a model across a variety of market conditions (SUDJIANTO; ZHANG, 2024).

These procedures are also essential in providing systematic assessments of model stability, including analysis under various market conditions, adaptability to simulated shocks and extreme events, and resilience during high-volatility periods. Such investigations could also include diagnostic tests for the stability of the coefficients and adaptability to meaningful changes in the data (ABOUZAID; BOUSSEDRA, 2025).

This process confirms that the models are dependable, coherent, and congruent with the purposes of this research. Hence, the evaluation provides the

necessary basis for the informed choice of the most adequate model, guiding the study's conclusion and its potential applications.

3.6. Deployment

Deployment is the phase in the CRISP-DM methodology which includes all activities to be performed in an operational environment, including packaging the code and integrating it into a system such that the organization can use it. It includes establishing real-time infrastructure and continually monitoring quality and effectiveness in a rapidly changing environment (SHIMAOKA et al., 2024).

3.6.1 Purpose of implementation in predictive modeling projects

This is the final and ongoing phase of the model lifecycle, whose aim is not only to ensure that developed models work technically but also to bring continuous value to the business (TRIPATHI et al., 2021). This involves converting predictions and insights into accessible and usable artifacts for end users, such as dashboards for analysts, visual notifications on existing systems, periodic reports to managers, or automated notifications to operations, ensuring that information is delivered in a manner that is understandable and actionable for various stakeholders (PODDAR et al., 2024).

3.6.2 Requirements for predictive model implementation

Conceptually, implementation requires observing minimum requirements on various fronts. This ranges from high-quality data, considering update frequency, formats, and sources, to an adequate infrastructure to ensure scalability and robustness, and well-established processes, such as data ingestion and validation pipelines, to ensure the model's reliability and continuous maintenance (TRIPATHI et al., 2021). For this implementation to be effective, it requires a multidisciplinary process that's tightly integrated with adopted operational systems, along with intuitive and useful user interfaces. This effort must encompass making predictions available for consultation, the precise

identification of responsible parties for every stage, and covers everything from updating the databases to disseminating the results (PODDAR et al., 2024).

3.6.3 Structure of a theoretical prediction pipeline

A theoretical prediction pipeline should involve a structured process that includes, but is not restricted to: Data ingestion and preprocessing, Feature engineering, Model building, Model evaluation, and the Persistence and publication of results, including standard components such as Extract, Transform, Load (ETL) and API deployment (RASCHKA et al., 2020; PODDAR et al., 2024).

Every step must apply best practices that enable the traceability and auditability of the process, such as mandatory log storage and change management, which provide greater transparency by allowing the monitoring of changes and model use (NOGARE; SILVEIRA, 2024). These logs record performance metrics, missing data, execution times and error events, and can be stored in systems (NOGARE; SILVEIRA, 2024).

Data versioning is fundamental for the reproducibility of experiments, reinforcing the importance of recording detailed information at every stage of the model lifecycle (PEIKERT; BRANDMAIER, 2021).

3.6.4 Operational data preparation

During implementation, the data preparation has an operational aspect that goes beyond what has been already done with transformations in the research setting. That is, there is a necessity to translate training queries into reusable pipelines that deliver data to the model, which may involve infrastructure for collecting data periodically or in real-time (PODDAR et al., 2024).

Preprocessing code, including unit tests, must automate basic checks like the validation of data completeness and consistency (PODDAR et al., 2024). Such validations may be carried out using specialized frameworks like Great Expectations or Deequ, which offer the possibility to define and document

expectations regarding the data, guaranteeing its integrity, accuracy, and consistency (DIAS et al., 2020).

3.6.5 Forecast generation: operational issues (Batch vs. Near Real-Time)

Predictions can be made in a variety of methods. Some of these strategies employ historical data and real-time alert information to anticipate the likelihood of future incident events, which leads to a Near-Real-Time operation mode where execution is performed upon new incoming data (REMIL et al., 2024).

In contrast, other approaches face with the issue of incident prediction and are usually made as predictions within a specific time interval, which suggests an operation mode closer to Batch or periodic executions (REMIL et al., 2024).

In the present work, the most feasible theoretical alternative would be Batch generation with a known horizon (t1, t6 and t12), along with validation routines that maintain consistency of the outputs before publication.

3.6.6 Results integration in the business context

The results integration must be planned based on the practical utility of forecasts for each target audience (SILVA; TIMO, 2022). From a conceptual perspective, the operationalization of a forecasting system is an issue related to clarify definitions about clear output formats, aggregation levels, and delivery mechanisms to ensure that the model results are usable by end-users (PODDAR et al., 2024). Tables and graphs summarizing forecasts are among the intended outputs of this study, which are aggregated by country, sector, and forecast horizon, and are made available through periodic reports and interactive dashboards.

User-related technical guidelines and specifications for results are of critical importance, as the effectiveness of forecasting software, in turn, depends on human perception and acceptance, particularly for non-experts. An opaque system can be rejected. Therefore, it is fundamental that the user can assess and interpret the forecasts for subsequent tasks (STROMER et al., 2024).

3.6.7 Theoretical monitoring and maintenance framework

The monitoring stage is essential to guarantee that models are not simply working technically, but also consistently delivering business value by adapting and optimizing based on new information and contextual changes through regular recalibration, periodic retraining, and continuous learning with new data (SUDJIANTO; ZHANG, 2024).

Triggers for the model reevaluation, but are not limited to, significant error increases evidenced by performance metrics, for example, degradation in RMSE, drift detection in data or concept that surfaces as altered input data distribution, and which might result in a compromised validity of the model, and changes parallel with a changing environment or market can bear direct implications on how data behave over time (SUDJIANTO; ZHANG, 2024).

3.6.8 Documentation, versioning, and reproducibility

Documentation and versioning are essential for traceability and transparency throughout the entire model lifecycle, allowing for the understanding of its operation and the audit of its decisions. Version control, for example, records changes to files over time so specific versions can be recalled later or revert the entire project to a previous state (PEIKERT; BRANDMAIER, 2021).

In relation to code versioning, systems such as Git and collaborative platforms like GitHub are used (SEMMELROCK et al., 2024). Additionally, versioning of data, artifacts and parameters is fundamental for reproducibility, being supported by domain-specific tools like DVC and MLflow, which are responsible for managing several artifacts associated with model performance and validation.

The possibility to easily revert any changes made is a key factor to foster experimentation and innovation, as it reduces the perceived risk involved in adopting new approaches (STROMER et al., 2024).

3.6.9 Academic reporting

Upon the completion of the implementation, an extensive academic report should be compiled to provide clear understanding for the entire process carried out, formalizing results and methodologies and insights achieved, and ensuring that generated knowledge is captured and disseminated for future applications and research (PODDAR et al., 2024).

In the context of this study, a paper in continuation of this monograph should include the applied CRISP-DM methodology, Random Forest and LightGBM model architecture and a detailed comparative analysis with ARIMA model results. To enable reproducibility, this report must include technical appendices, such as code segments, samples of transformed data, and information about execution environments.

A consideration of the limitations of the study, opportunities for future research and the ethical and regulatory implications of utilizing advanced predictive models in an economic context must also be included (SUDJIANTO; ZHANG, 2024).

4 TIMELINE AND SCHEDULE

The timeline below presents the organization of project activities over time, distributed between August 2025 and June 2026. It includes all stages of development, from initial conception to delivery and defense of the final project, demonstrating the logical sequence and time required for each phase.

Activities	Aug/25	Sep/25	Oct/25	Nov/25	Dec/25	Jan/26	Feb/26	Mar/26	Apr/26	May/26	Jun/26
Project Topic Selection	X										
Project Structure Development	X	X									
Development of Objectives, Theme Delimitation, Problem Definition	X	X									
Selection and Reading of Works	X	X	X	X							
Formulation of Justification and Contextualization	X	X									
Definition of Hypotheses or Research Questions	X	X									
Drafting the Monograph	X	X	X	X							
Final revision of the Monograph				X							
Submission of the Monograph				X							
Data Collection					X	X					
Data Preparation						X	X	X			
Modeling and Testing							X	X	X		
Evaluation of Results									X		
Interview with Experts for Confirmation									X	X	
Consolidation/Interpretation of Findings										X	
Final revision of the work											X
Project submission											X
Thesis defense											X

5 APPENDIX

5.1 Interview script

In order to validate the results obtained and minimize bias in the research, open-ended questions were designed for two distinct profiles: professionals with extensive knowledge on foreign trade and international business, and IT professionals. The questions had been worded in an unbiased and open way specifically not to induce responses so that the participants could communicate generally and in detail.

The script aims at capturing these opinions to add a qualitative validation of the models and methodological choices used in this work. The questions address the practical applicability of the forecasts, the relevance of the factors chosen, the acceptance of the models, and the consistency of the decisions adopted in the project.

5.1.1 Questionnaire for foreign trade / international business professionals

How do machine learning forecasting models, e.g., Random Forest, LightGBM, compare with traditional time-series models such as ARIMA for the task of forecasting in foreign trade? What do you perceive are the strengths or limitations of each approach?

Are economic factors considered in the models (exchange rate, inflation, GDP, and other variables) relevant to predict international trade indicators? Are there other factors that you might be inclined to include?

How might the outcomes anticipated by these models be valuable in decision-making within your field? Can you provide some examples of practical applications, e.g., export/import planning or demand analysis, where these forecasts would have an impact?

What is your sense of the interpretability and trust that managers and analysts might have in these predictive models? Would you trust a classic

statistical model (ARIMA) or a machine learning model (Random Forest, LightGBM) more? Why?

What other fail-safe checks or additional safety measures would you recommend to make sure that such models do not introduce unwanted biases in the analysis of foreign trade?

5.1.2 Questionnaire for technology professionals

Technically speaking, what are the main challenges and benefits of utilizing machine learning models (Random Forest, LightGBM etc.) for forecasting economic or business indicators compared to traditional time series models such as ARIMA? And which approach has proven more effective according to your experience?

In the sense of data and variables, do you think the chosen economic factors, e.g., macroeconomic variables, are suitable to feed predictive models? Do you think other types of data should be included, e.g., sectoral indicators, market data, or alternative sources, to make it more accurate?

What explainability or visualization strategies do you consider important for communicating the results of complex models to a non-technical audience? How can transparency be ensured in the modeling process so that the model's decisions are understandable?

How do you feel the technology community, or IT companies, generally perceives the acceptance of ML-based predictive models compared to traditional statistical approaches? What metrics or technical evidence would be necessary to convince stakeholders to adopt an ML model?

Have you deployed predictions by your model, either ARIMA or ML, into a live system/application? What was that experience like? Many of the descriptors shared for this were a variation of "I can work with an acceptable model." What made the model acceptable, e.g., ease of integration, production performance, and user confidence?

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